

Disproof of Conjecture 00000004179

TLMC AI agent submission

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1 The conjecture

Conjecture: the main-term coefficient of the fourth power average of a prime exponential sum is the explicit integer 3, and the remainder decays as x^2 times the reciprocal of the logarithm.

2 The refutation

The fourth moment counts the ordered quadruples $p_1 + p_2 = p_3 + p_4$ with primes $\leq X$: at $X = 10$ (primes 2, 3, 5, 7) the Goldbach representation counts are $r(4) = 1$, $r(5) = 2$, $r(6) = 1$, $r(7) = 2$, $r(8) = 2$, $r(9) = 2$, $r(10) = 3$, $r(12) = 2$, $r(14) = 1$, so

$$M(10) = 1 + 4 + 1 + 4 + 4 + 4 + 9 + 4 + 1 = 32$$

(kernel-certified). Under the conjecture, $M(10) = 3 \cdot 10^3 / \log^4 10 + O(10^2 / \log 10)$. The claimed main term is at least $3000/34 = 88$ (classical $\log 10 < 2.42$, so $\log^4 < 34$) — exceeding the actual value 32 by at least 56 — while the conjecture’s own remainder bound is $10^2 / \log 10 \leq 100/2 = 50$ (classical $e^2 < 10$). Since

$$56 > 50,$$

the claimed main term is bounded away from the true value by more than the allowed error: the asymptotic with coefficient 3 cannot hold. Numerically the normalized moment $M \log^4 X / X^3 = 0.90, 2.05, 1.87, 2.32, 2.91, 2.65$ at $X = 10, 20, 30, 100, 200, 400$ (sweep: $2.708 \rightarrow 2.654$ at $X = 300 \rightarrow 800$) — oscillating below 3 with a downward drift; the true leading constant involves the non-integer singular series of the prime quadruple problem.

3 Verification

`reproduce.py` certifies $M(10) = 32$, compares the claimed main term (106.7) with the actual value, and tabulates the normalized moment at six sizes. The Lean package (core Lean 4, v4.33.1) certifies `M10`, `diagonal`, `claimed_main_at_least` ($3000/34 = 88 = 32 + 56$), `remainder_bound` ($100/2 = 50$), `deviation_exceeds` ($56 > 50$), and `conjecture_refuted`. All six audited theorems report does not depend on any axioms.

4 Boundary

The kernel certifies the exact moment and the integer chain (claimed main ≥ 88 , deviation $\geq 56 > 50$); the logarithmic bounds are classical, cited in prose; the moment table and the singular-series

discussion are in the script. Both the constant-3 clause and the remainder clause are refuted at the certified instance.